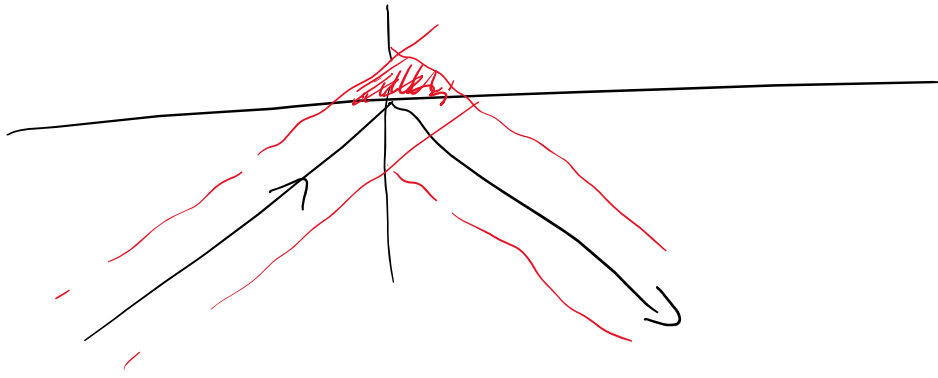




- evanescent wave is stationary $\rightarrow$ not propagating
 $\hookrightarrow$ google standing wave.

- K becomes imaginary

$$K = j\beta$$

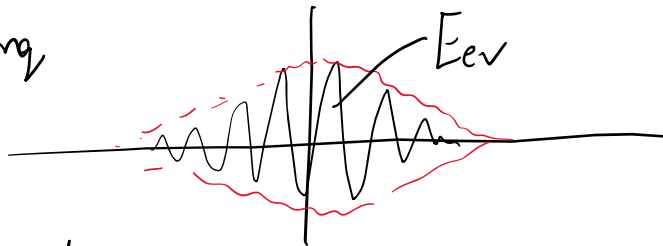
standard $E = E_0 e^{-j(\omega t - Kx)}$

$$E = E_0 e^{-j(\omega t - j\beta x)}$$

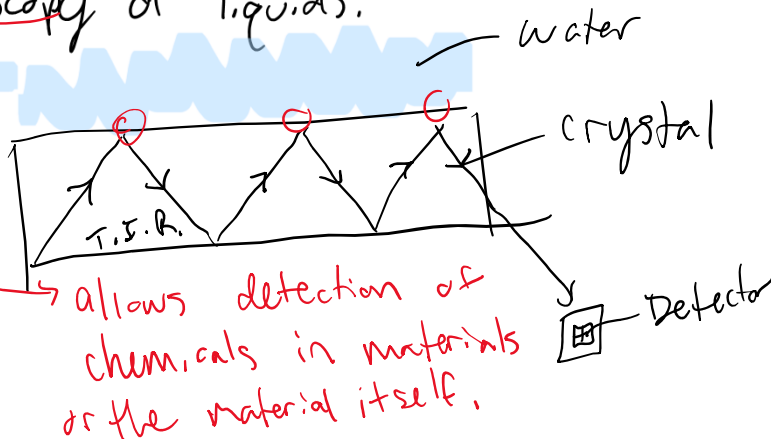
$$E_{ev} = E_0 e^{-j\omega t} (e^{-\beta x})$$

Real term

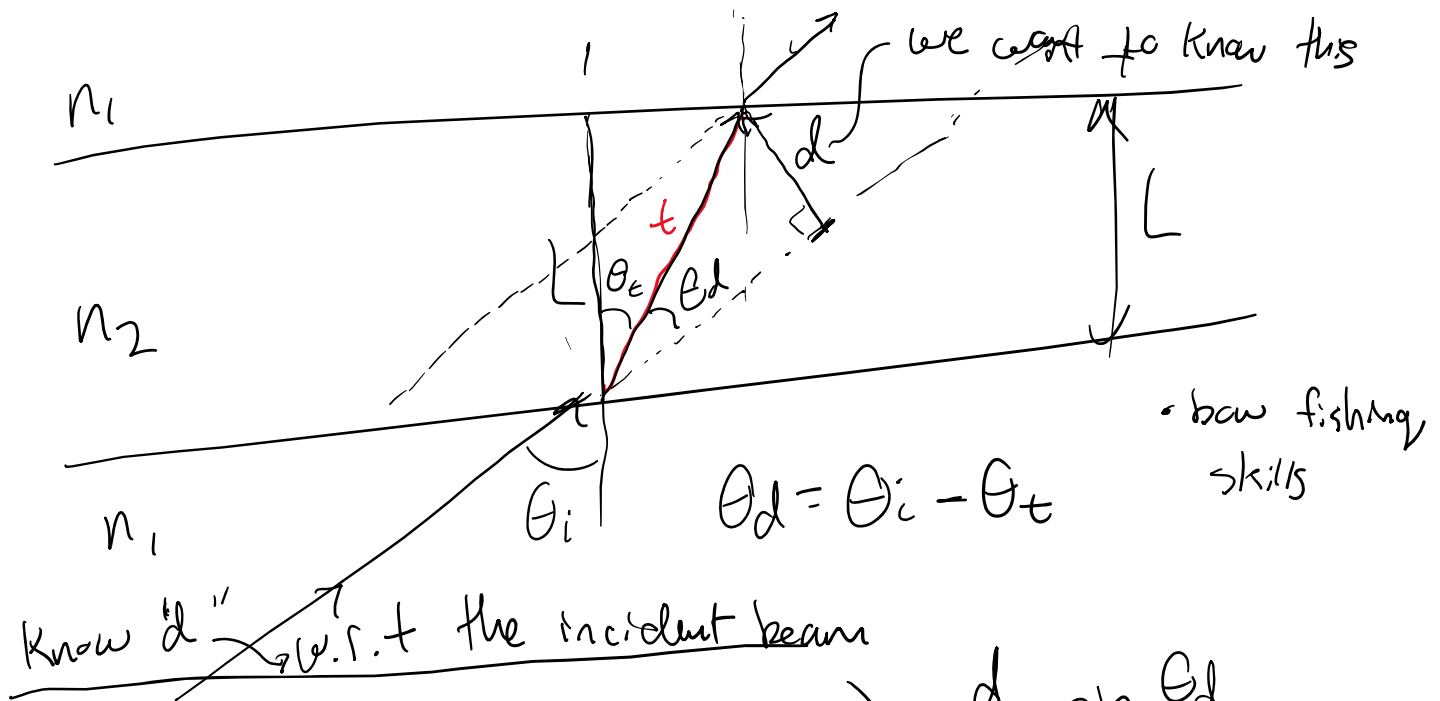
- Produced by radiating waves in a material



- Spectroscopy of liquids.



- Microscopes that use fraction of wavelengths, not limited by the full wavelength



$\theta_i < \theta_c$ (Do Not have T.I.R.)

$$\frac{d}{t} = \sin \theta_d$$

$$\frac{d}{t} = \sin(\theta_i - \theta_t)$$

$$d = t \sin(\theta_i - \theta_t)$$

$$\frac{L}{t} = \cos(\theta_t)$$

$$t = \frac{L}{\cos(\theta_t)}$$

$$d = L \left[\frac{\sin(\theta_i - \theta_t)}{\cos(\theta_t)} \right]$$

Trig ID.

$$\sin(\theta_i - \theta_t) = \sin(\theta_i) \cos(\theta_t) - \cos(\theta_i) \sin(\theta_t)$$

$$\text{also } \cos(\theta_t) = \sqrt{1 - \sin^2 \theta_t}$$

$$\frac{d}{L} = \left[\frac{\sin(\theta_i) \cos(\theta_t) - \cos(\theta_i) \sin(\theta_t)}{\sqrt{1 - \sin^2 \theta_t}} \right]$$

$$\frac{d}{L} = \left[\frac{\sin(\theta_i) \cos(\theta_t) - \cos(\theta_i) \sin(\theta_t)}{\sqrt{1 - \sin^2 \theta_t}} \right]$$

Snell's Law

$$n_t \sin \theta_t = n_i \sin \theta_i \rightarrow \text{can be also in terms of cosines.}$$

$$(\sin \theta_i) (\cos \theta_t) - \frac{n_i}{n_t} (\sin \theta_i) (\cos \theta_i)$$

$$\frac{d}{L} = (\sin \theta_i) \left[1 - \frac{\cos \theta_i}{\sqrt{\left(\frac{n_t}{n_i}\right)^2 - \sin^2 \theta_i}} \right]$$

Before Next test

if you can show this step $\rightarrow$ you get +5 on next test